

White Paper: A Pathway to a Macroscopic Stable Light Spheroid via Solid-State Microresonators

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We present an engineering concept for a macroscopic Stable Light Spheroid (SLS) – a long-lived, localized volume of confined light with record energy density. Overcoming the fundamental limitations of ultracold atom platforms, our architecture leverages the integration of solid-state Bragg microresonators pumped by coherent laser diode arrays. This work provides a systematic decomposition of instability, microscopic scale, and high energy consumption problems, followed by synthesis of a solution from nanophotonics and semiconductor technologies. We establish rigorous quantitative verification criteria, with the key metric being a confined light lifetime of $\tau > 1$ s, exceeding the free-flight time by 11 orders of magnitude. This approach transforms the challenge from fundamental physics to multidisciplinary engineering integration.

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I. INTRODUCTION

Attempts to create stable, macroscopic, long-lived light structures have traditionally faced insurmountable challenges in platforms based on ultracold atoms and quantum dots [1, 2].

The high sensitivity of magnetic and optical traps to vibrations and noise, fundamental size limitations of atomic condensates (< 100 μm), and enormous energy consumption of cryogenic systems render these approaches impractical for macroscopic object realization.

This work proposes a paradigm shift: abandoning fragile atomic systems in favor of an *engineering-oriented platform* based on solid-state photonic microresonators. We discover

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no new physical phenomena but conduct systematic engineering analysis and synthesis, assembling the setup from known, well-established technologies.

Our contribution is threefold:

1. **Decomposition** of fundamental problems in existing approaches.
2. **Synthesis** of a new architecture based on solid-state microresonators.
3. **Definition** of rigorous quantitative experimental verification criteria.

II. PROBLEM DECOMPOSITION

A. Instability of Atomic Platforms

The problem is compound:

- **(a) External vibrations:** Magnetic traps require peta- and femto-tesla stability, unattainable in laboratory conditions.
- **(b) Field fluctuations:** Power supply drift and interference destroy condensates within milliseconds.
- **(c) Atom collisions:** Vacuum imperfection and inelastic collisions limit lifetime.

B. Microscopic Scale

Size limitation stems from:

- Impossibility of creating large (> 1 mm) homogeneous ultracold atom clouds.
- Inhomogeneous pumping of large volumes with single laser beams.

C. Energy Inefficiency

Over 99% of energy is wasted:

- Cryogenic systems for microkelvin temperatures.
- Low efficiency ($< 10\%$) of gas and solid-state pump lasers.

III. ARCHITECTURE SYNTHESIS

The proposed architecture directly addresses the decomposed problems.

A. Active Medium: Solid-State Microresonator

We propose using semiconductor microresonators with distributed Bragg reflectors (DBRs) as the active medium [3]. This structure:

- **Solves (a), (b):** Crystal lattice is inherently stable. No magnetic traps required.
- **Solves scale problem:** Lithography enables resonators with mode volumes $V \sim 1 \text{ mm}^3$.
- **Creates polaritons:** Strong exciton-photon coupling forms Bose-Einstein condensate of polaritons [4] at 4-150 K temperatures.

The SLS shape is defined by resonator geometry (e.g., plano-concave).

B. Pumping System: Coherent Laser Array

For homogeneous large-volume pumping, we propose using arrays of synchronized single-mode laser diodes.

- **Solves inhomogeneity:** Uniform illumination of the entire surface.
- **Solves efficiency:** Modern diodes have $> 60\%$ wall-plug efficiency.

C. Cooling and Stabilization

The system operates in a compact cryocooler at 20-80 K.

- **Reduces energy consumption** by 3 orders compared to atomic systems.
- Suppresses phonon excitations, increasing coherence time.

IV. QUANTITATIVE SUCCESS CRITERIA

We propose the following verification protocol.

A. Parameter: Energy Density

- **Target:** $u > 2000 \text{ J/m}^3$.
- **Formula:** $u = \frac{E}{V} = \frac{P \cdot \tau}{V}$.
- **Calculation:** For $P = 1 \text{ mW}$, $\tau = 1 \text{ s}$, $V = 5.2 \times 10^{-7} \text{ m}^3$ ($D = 1 \text{ cm}$): $u \approx 1900 \text{ J/m}^3$.

B. Criterion 1: Lifetime (Key Metric)

- **Target:** $\tau > 1 \text{ s}$.
- **Methodology:** Abrupt pump shutdown, recording intensity decay $I(t)$.
- **Verification:** Fitting $I(t) = I_0 \exp(-t/\tau)$ yields $\tau > 1 \text{ s}$.
- **Physical meaning:** Exceeds free-flight time ($\sim 30 \text{ ps}$) by $> 10^{11}$ times.

C. Criterion 2: Spatial Stability

- **Target:** Boundary position standard deviation $\sigma < 10 \text{ }\mu\text{m}$.
- **Methodology:** Imaging with ICCD camera.

D. Criterion 3: Controllability

- **Target:** Controlled displacement $\Delta x > D$ via external field (electric, thermal).
- **Methodology:** Gradient application, position tracking.

V. DISCUSSION AND CONCLUSION

We present a complete engineering concept demonstrating no fundamental prohibitions for SLS creation. Transition to solid-state platform removes key limitations of atomic systems.

- **Technology readiness:** All components (DBR resonators, laser arrays, cryocoolers) have TRL 7-9.
- **Interdisciplinary nature:** Requires collaboration between nanophotonics, semiconductor physics, and cryogenics specialists.
- **Verifiability:** Proposed criteria unambiguously prove experimental success.

This work opens the pathway to practical realization of stable macroscopic light objects with applications ranging from quantum memory to novel radiation sources.

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